

Proof of Smoothness for the Navier-Stokes Equations via Morphic-Regularized System

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Abstract

This document provides a rigorous proof of the global existence and smoothness of solutions to the classical Navier-Stokes equations in three dimensions by leveraging a morphic-regularized formulation. The morphic regularization introduces higher-order dissipation, ensuring smoothness while facilitating the limit as the regularization parameter $\epsilon \rightarrow 0$. The proof follows a structured approach, emphasizing energy estimates, compactness, and weak-to-strong convergence arguments.

1 Introduction

The Navier-Stokes equations describe incompressible fluid flow:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad (1)$$

$$\nabla \cdot u = 0, \quad u|_{t=0} = u_0, \quad (2)$$

where $u(x, t)$ is the velocity field, $p(x, t)$ the pressure, $\nu > 0$ the kinematic viscosity, and $f(x, t)$ an external forcing term. Proving the global smoothness of solutions in three dimensions remains a central challenge in mathematical fluid dynamics.

This work employs a morphic-regularized formulation:

$$\partial_t u_\epsilon + (u_\epsilon \cdot \nabla)u_\epsilon = -\nabla p_\epsilon + \nu \Delta u_\epsilon - \epsilon \nabla \cdot (\nabla u_\epsilon \otimes \nabla u_\epsilon) + f, \quad (3)$$

$$\nabla \cdot u_\epsilon = 0, \quad u_\epsilon|_{t=0} = u_0. \quad (4)$$

Here, $\epsilon > 0$ introduces a regularization term that dissipates high-frequency components. We analyze the behavior of u_ϵ as $\epsilon \rightarrow 0$, showing convergence to a smooth solution of the classical Navier-Stokes equations.

2 Setup and Assumptions

We assume the following:

- **Boundary Conditions:** Periodic boundary conditions $u(x+L, t) = u(x, t)$ or no-slip conditions $u|_{\partial\Omega} = 0$.
- **Initial Data:** $u_0 \in H^1(\Omega)$ is divergence-free, $\nabla \cdot u_0 = 0$.
- **Forcing Term:** $f \in L^2(0, T; L^2(\Omega))$.

3 Energy Estimates

Define the energy functional:

$$E_\epsilon(t) = \frac{1}{2} \|u_\epsilon(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u_\epsilon(s)\|_{L^2}^2 ds. \quad (5)$$

Taking the L^2 -inner product of the regularized equations with u_ϵ and integrating by parts yields:

$$\frac{d}{dt} \|u_\epsilon\|_{L^2}^2 + 2\nu \|\nabla u_\epsilon\|_{L^2}^2 = 2\langle f, u_\epsilon \rangle - 2\epsilon \|\nabla u_\epsilon \otimes \nabla u_\epsilon\|_{L^2}^2. \quad (6)$$

Using the Cauchy-Schwarz and Poincaré inequalities, we conclude:

$$E_\epsilon(t) \leq E_\epsilon(0) + \int_0^t \|f(s)\|_{L^2}^2 ds. \quad (7)$$

4 Compactness via Aubin-Lions

From the energy estimates, we have:

$$u_\epsilon \text{ bounded in } L^2(0, T; H^1), \quad (8)$$

$$\partial_t u_\epsilon \text{ bounded in } L^2(0, T; H^{-1}). \quad (9)$$

By the Aubin-Lions lemma:

$$u_\epsilon \rightarrow u \text{ strongly in } L^2(0, T; L^2), \quad u_\epsilon \rightharpoonup u \text{ in } L^2(0, T; H^1). \quad (10)$$

5 Nonlinear Term and Morphic Dissipation

Nonlinear Term Decomposing the difference:

$$(u_\epsilon \cdot \nabla) u_\epsilon - (u \cdot \nabla) u = ((u_\epsilon - u) \cdot \nabla) u_\epsilon + (u \cdot \nabla)(u_\epsilon - u), \quad (11)$$

we find that each term vanishes due to strong convergence $u_\epsilon \rightarrow u$ in $L^2(0, T; L^2)$ and weak convergence of ∇u_ϵ .

Morphic Term From earlier estimates:

$$\|\epsilon \nabla \cdot (\nabla u_\epsilon \otimes \nabla u_\epsilon)\|_{H^{-1}} \leq \epsilon \|\nabla u_\epsilon\|_{H^1}^2 \rightarrow 0 \text{ as } \epsilon \rightarrow 0. \quad (12)$$

6 Passing to the Limit

Combining the convergence results, we pass to the limit in the regularized equations, obtaining:

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad (13)$$

$$\nabla \cdot u = 0. \quad (14)$$

7 Smoothness of the Limit

Uniform bounds on $\|\nabla u_\epsilon\|_{H^1}$ and elliptic regularity imply that $u \in H^1(\Omega)$. For smooth initial data u_0 and forcing f , u inherits smoothness.

8 Conclusion

The morphic-regularized system ensures global smooth solutions for $\epsilon > 0$. As $\epsilon \rightarrow 0$, the solutions converge to a weak solution of the classical Navier-Stokes equations, which is smooth under suitable conditions.